Switchable CPU Model -NXP VDKs S32NXX, S32KXXX, S32K5X

User Guide

Version: V1.0.0

Synopsys
8/7/25

Introduction:

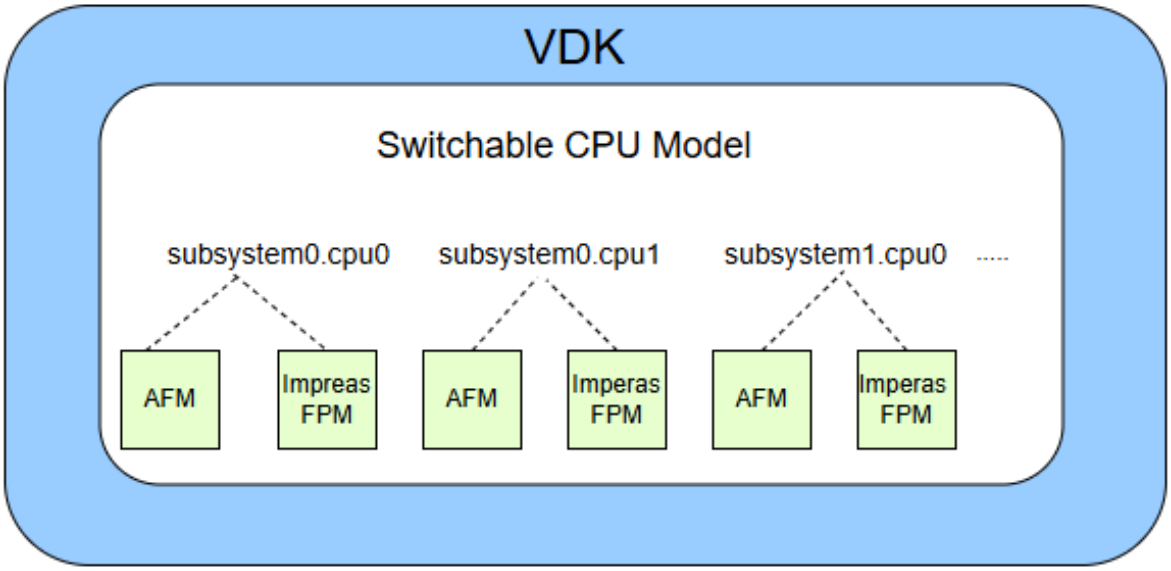
A switchable CPU model is a simulation model that enables dynamic switching between different configurations at runtime. Instead of maintaining two separate VDKs for the ARM Fast Models (AFM) and the Fast Processor Model (ImperasFPM), we have integrated the switchable CPU model that provides the flexibility to select the CPU variant as needed using vpconfig parameter.

The following Switchable CPU models are supported:

1. R52 CPU Model
2. Cortex M7 CPU Model

Block Diagram:

VDK switchable CPU model Block Diagram:



How to Switch:

This CPU model is currently based on the ARM core architecture. The ARM Core can be switched between AFM or ImperasFPM core functionality by configuring the variant parameter.

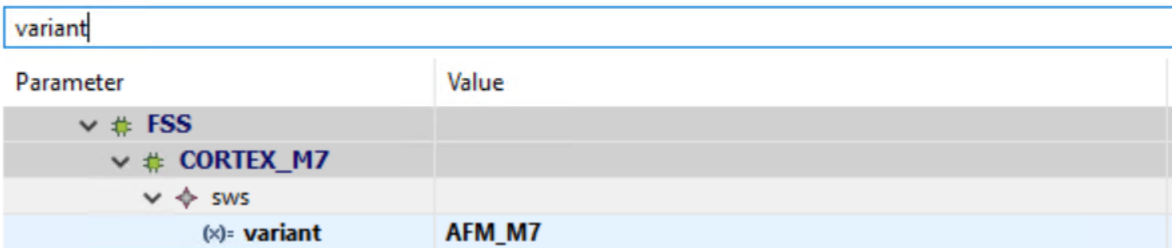
This parameter will help user to select the variant of CPU model, allowing the VDK to adapt its initialization and behaviour based on the chosen variant.

Default variant is Fast Model (AFM) in all the Switchable CPU models – however, once integrated in the VDK, this default can be changed to Impers or QEMU variant once the VDK has been fully tested with our system tests and with customer software and there is good confidence level on the faster CPU model.

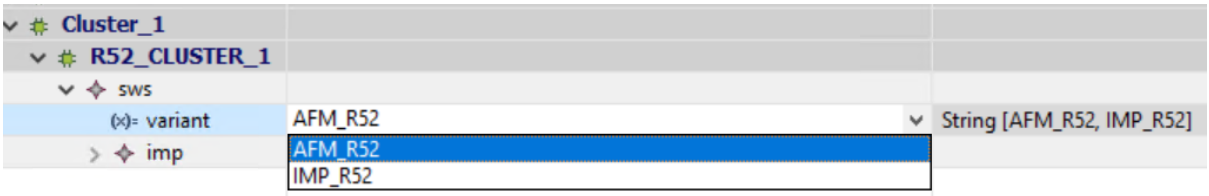
Steps to switch from ARM mode to ImperasFPM variant:

Note: *Refer below steps as an example. The same steps would be applicable for all NXP VDKs.*

1. Open vpconfig file (. Vpcfg) file in software skin e.g., Dhrystone-FSS_core0.vpcfg.
2. Search for variant parameter as shown below.



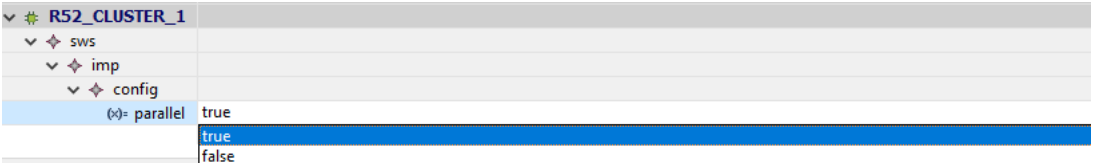
3. Change the value of the 'variant' parameter from AFM_R52 to IMP_R52 (or from AFM_M7 to IMP_M7 for the M7 case) as shown below. AFM refers to Arm fast models and IMP refers to Imperas Fast Processor Models.



4. Save the configuration file and run vp config to apply changes.

Parallel Mode

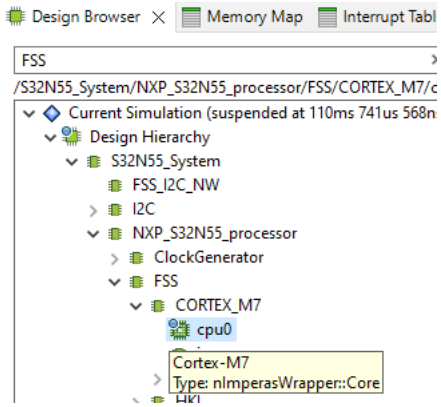
- In ImperasFPM CPU model supports parallel mode (multi thread) can be enabled by setting as true as shown below.



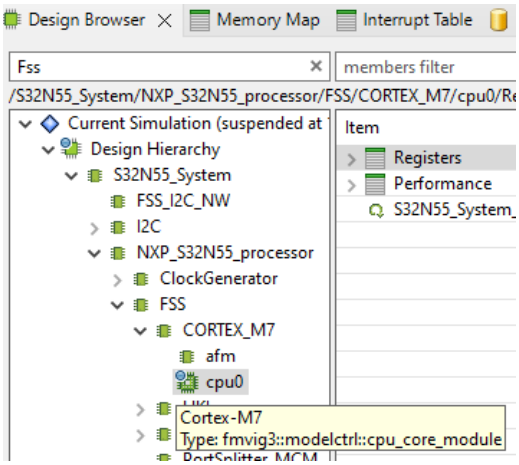
Note: The parallel mode(multi-threaded) is still in its early stages. The software, which works seamlessly in sequential mode, may encounters unexpected behaviour or failures in parallel mode. This needs to be debugged jointly on case-by-case basis. Parallel mode introduces complexities in thread synchronization and resource sharing, which may lead to race conditions or deadlocks. These issues are unlikely to be present in sequential mode, where execution is more predictable. Specific software components may not have been thoroughly tested with parallel mode enabled, leading to compatibility issues. This is particularly critical for applications that rely on precise timing or deterministic execution. **The recommendation is to keep the parallel model switched off by default. This can be enabled for specific SW/applications and use cases.**

- User can confirm the current CPU model selected from Design Browser by hovering the mouse over CPU to see variant as shown below.

  In case of ImperasFPM:



In case of AFM:



Known Limitations:

Imperas FPM R52:

1. **Stall command** is not supported by ImperasFPM R52 for now, VDK user needs to clamp the reset pin for stalling the cores.

   Example:
   sim.BoolProbe("/S32Z270_System/S32Z270_MCU/RTU_0/R52_CLUSTER_0/nCORERESET_1_pin").set_clamp_value (0)
2. ***spsr_hyp*** register is not supported by ImperasFPM R52 core, users need to update the SW with different instruction called spsr_cxsf.
3. GDB ports are not supported for ImperasFPM R52 cores
4. ***SNPS_OP_LDR_USE_PHYSICAL*** environment variable needs to be set to 1 when the elf file specifies both virtual and physical addresses
5. Mismatch in core register names between AFM and ImperasFPM

Imperas FPM CM7:

1. Mismatch in core register names between AFM and ImperasFPM